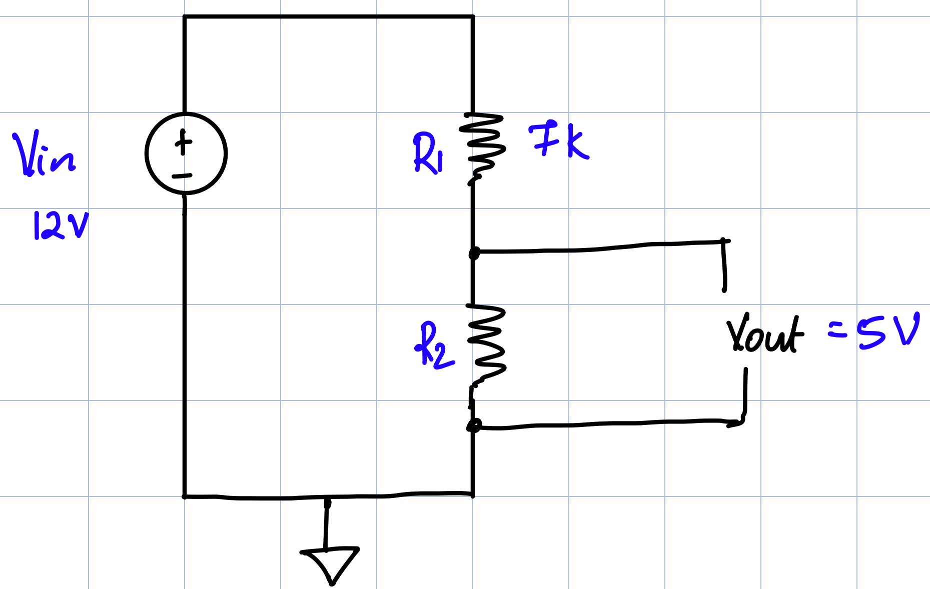
EECE 5685

Virtual Lab #1: Linear Regulator

Part 1: Resistor Divider



a) given $V_{in} = 12V$

$R_1 = 7K$

$V_{out} = 5V$

We know that for a resistor divider circuit $V_{out} = \dfrac{V_{in} R_2}{R_1 + R_2}$

$$\therefore V_{out} = \frac{12 \times R_2}{7 + R_2}$$

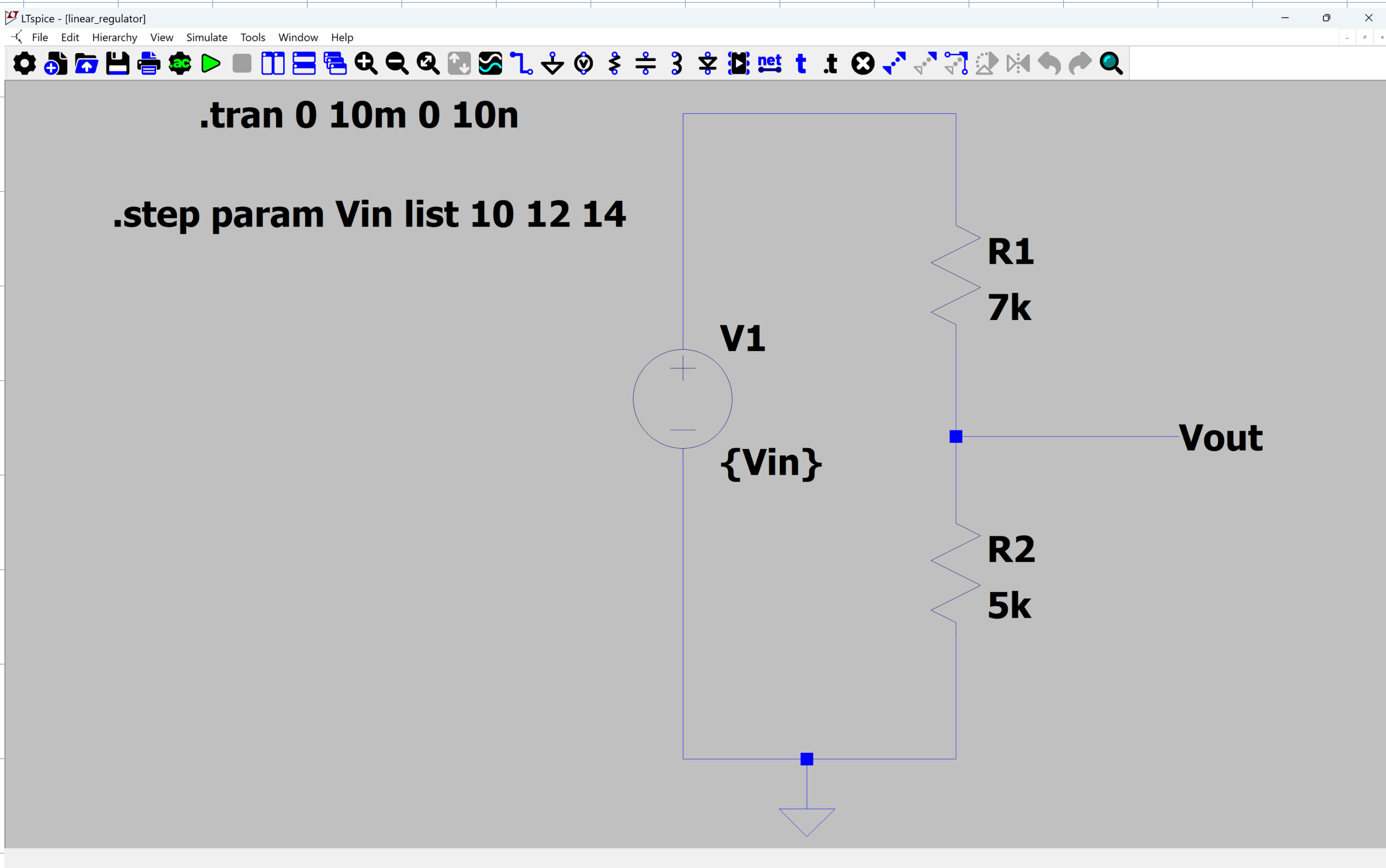
$$5 = \frac{12R_2}{7+R_2}$$

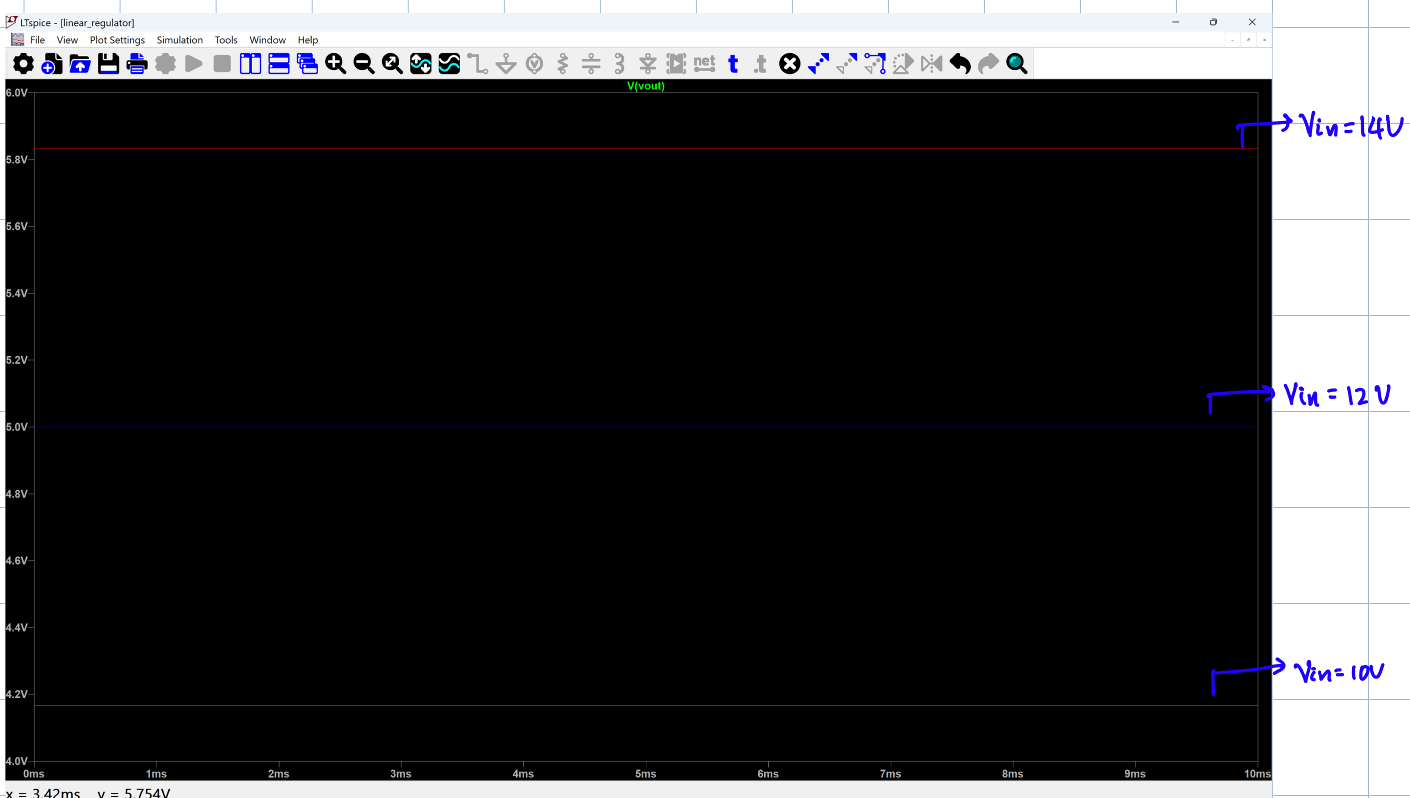
$$35 + 5R_2 = 12R_2$$

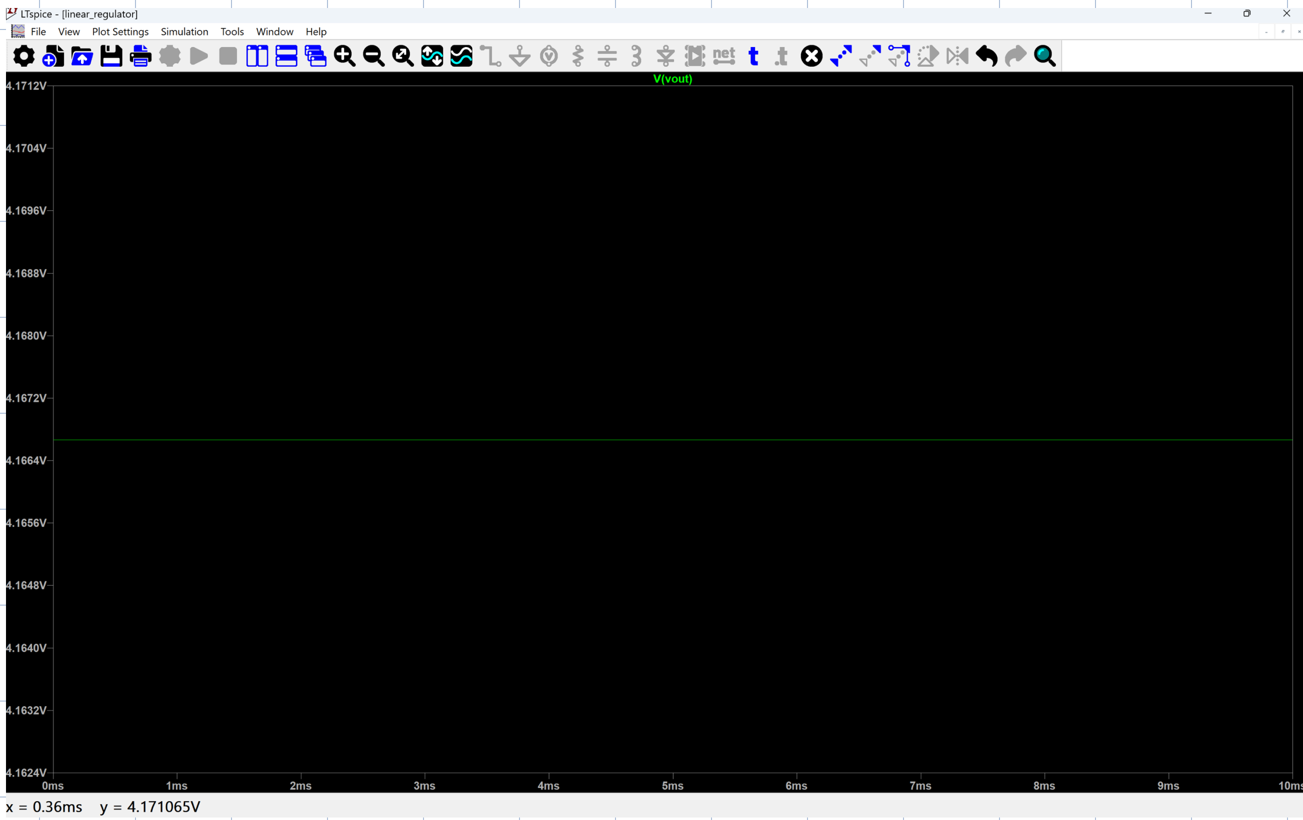
$7R_2 = 35$

$R_2 = 35/7$

$\boxed{R_2 = 5k\Omega}$

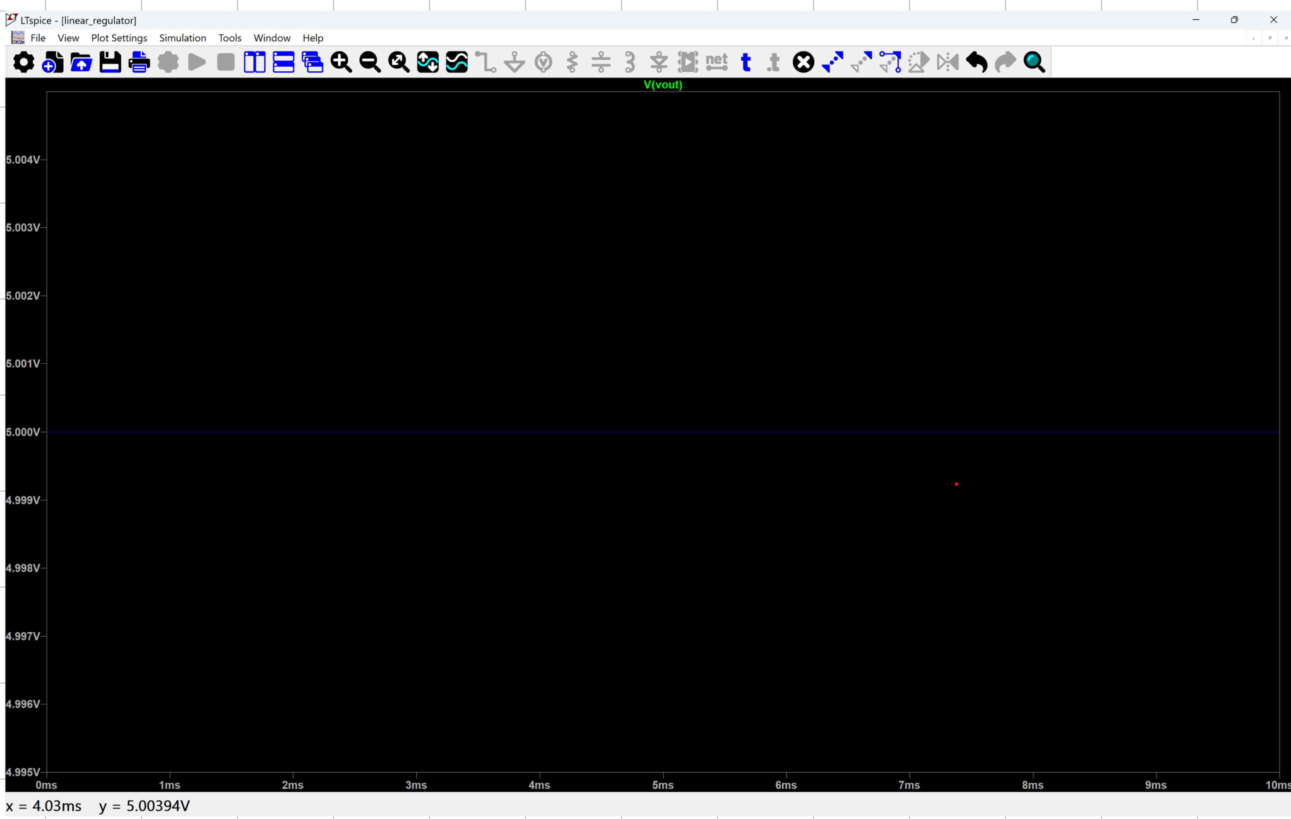






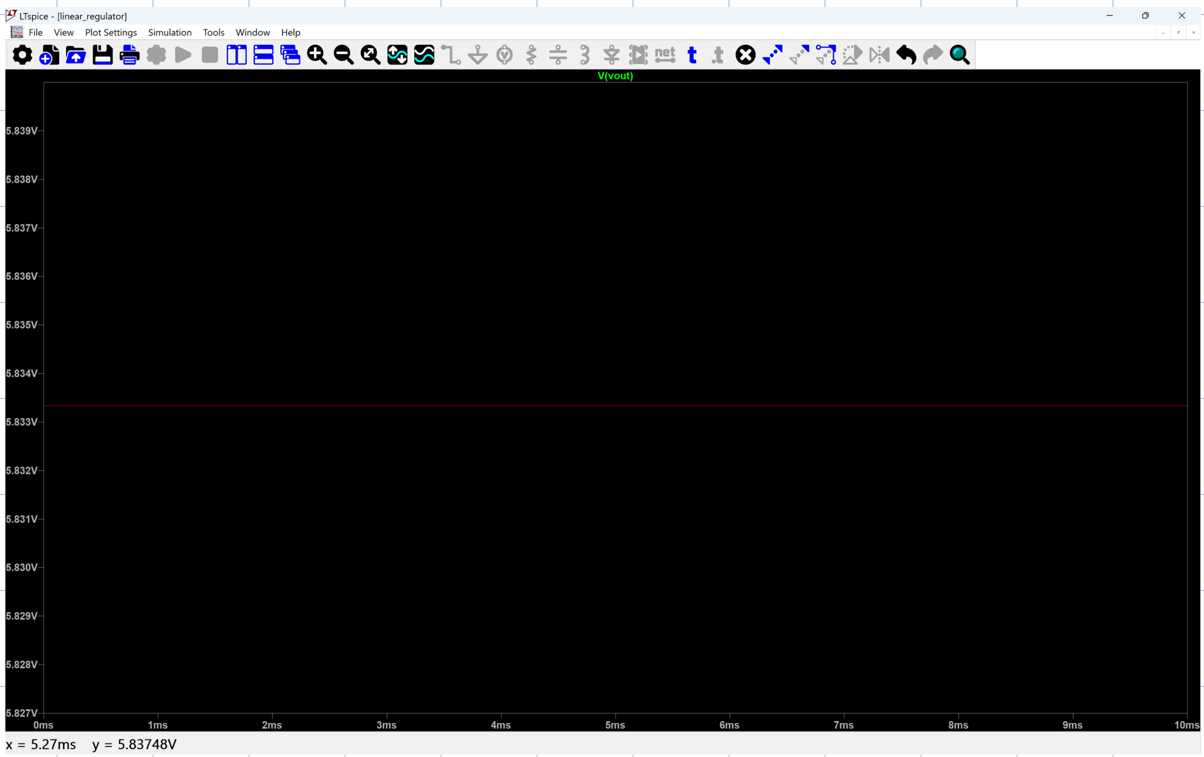
$V_{in} = 10V$

$V_{out}: 4.166V$



$V_{in} = 12V$

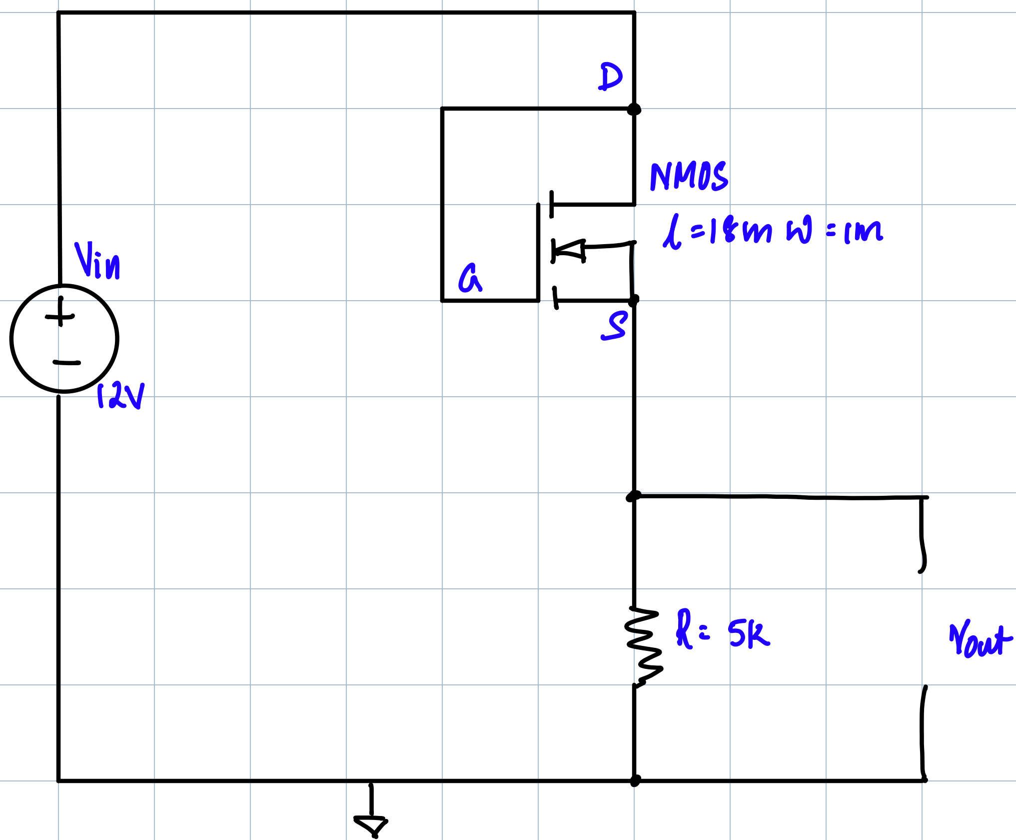
$V_{out} = 5V$



$V_{in} = 14V$

$V_{out} = 5.833V$

Part II: Linear Regulator Using a MOSFET (n-mos4)



d)

Given: $V_{in} = 12V$

$R = 5k\Omega$

NMOS:

a) $V_T = 1V$

b) $k'_n = 1mA/V^2$

c) $W = 1mm$

d) $L = 18mm$

To find V_{GD}, V_{DS} and V_{out}

Solution: Since the gate is shorted to the drain, $V_D = V_G$ [short circuit current is zero, voltages at drain and gate are same)

$V_{GS} = V_G - V_S$

$V_{DS} = V_D - V_S$

Since $V_{DS} = V_G - V_S$

$\boxed{V_{DS} = V_{GS}}$

Since R_1 and R_2 are connected in series the current through R_1 and R_2 are the same.

$\therefore$ I_D the drain current will be the same current through R_2 i.e I_{R_2}

The voltage across resistor R_2 $= V_{out} = I_{R_2} R_2$

$$V_{out} = I_D R_2$$

$$I_D = \frac{V_{out}}{R_2}.$$

$$\boxed{I_D = \frac{V_{out}}{5000}} \quad \because R_2 = 5k\Omega$$

We know that the drain current for a Mosfet is given by

$$I_D = \frac{k_n'}{2}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2 \quad \rightarrow ①$$

Substituting $I_D = \frac{V_{out}}{5000}$

We know that according to KVL $V_{in} = V_{DS} + V_{out}$

$$V_{out} = V_{in} - V_{DS}$$

substituting in ①

$$\frac{V_{in} - V_{DS}}{5000} = \frac{10^{-3}}{2} \times \frac{1}{18} (V_{GS} - 1)^2$$

Since $V_{GS} = V_{DS}$ ($\because$ Drain and gate are shorted)

$$12 - V_{DS} = \frac{5000\times10^{-3}}{2} \times \frac{1}{18} (V_{DS}-1)^2$$

$$12 - V_{DS} = \frac{5}{36} (V_{DS}-1)^2$$

$$432 - 36V_{DS} = 5(V_{DS}^2 + 1 - 2V_{DS})$$

$$432 - 38V_{DS} = 5V_{DS}^2 + 5 - 10V_{DS}$$

$$5V_{DS}^2 + 5 - 10V_{DS} - 432 + 36V_{DS} = 0$$

$$5V_{DS}^2 + 26V_{DS} - 427 = 0$$

$$\boxed{\therefore V_{DS} = -\frac{61}{5}, 7}$$

$V_{DS} = 7V$

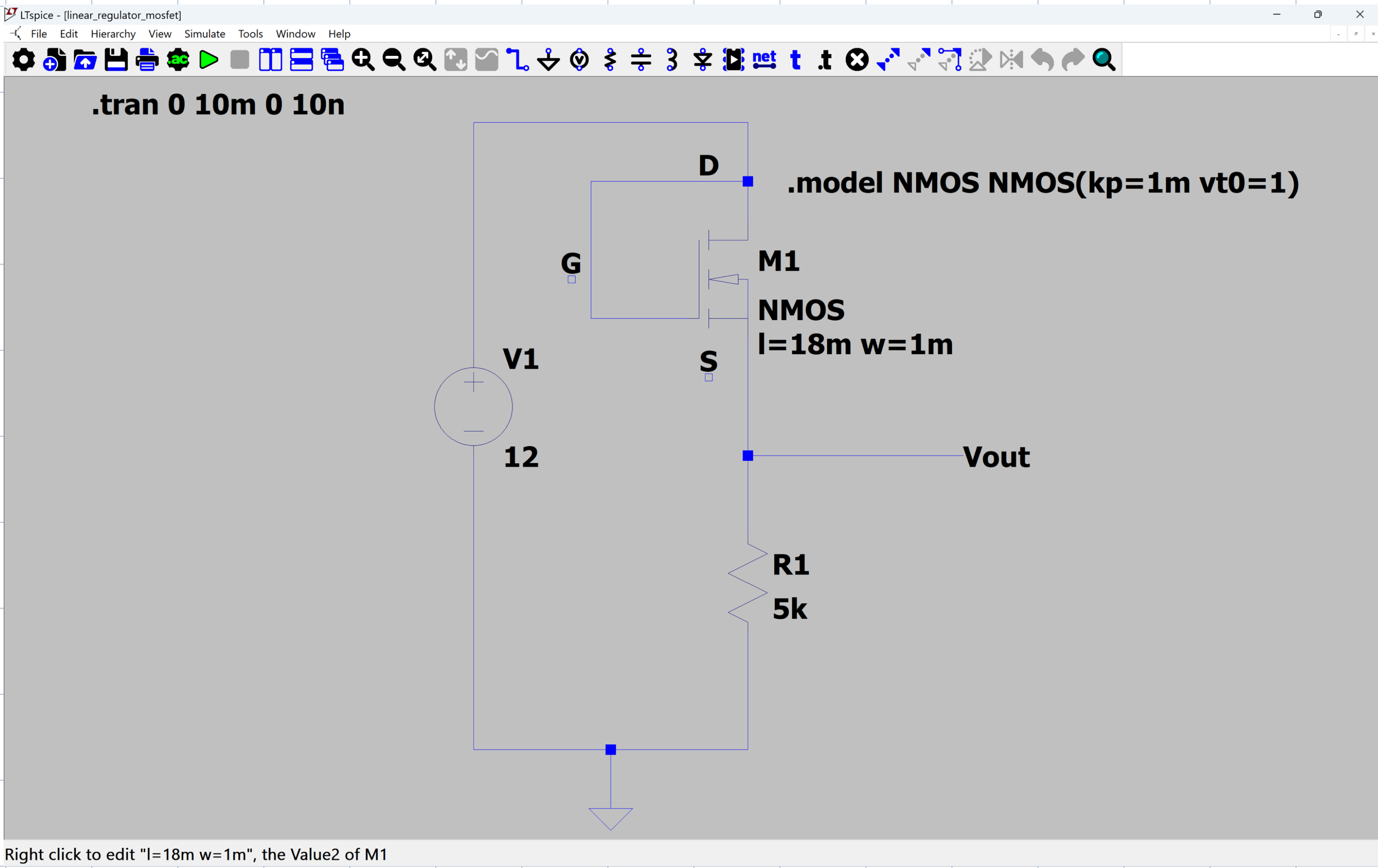
Sine

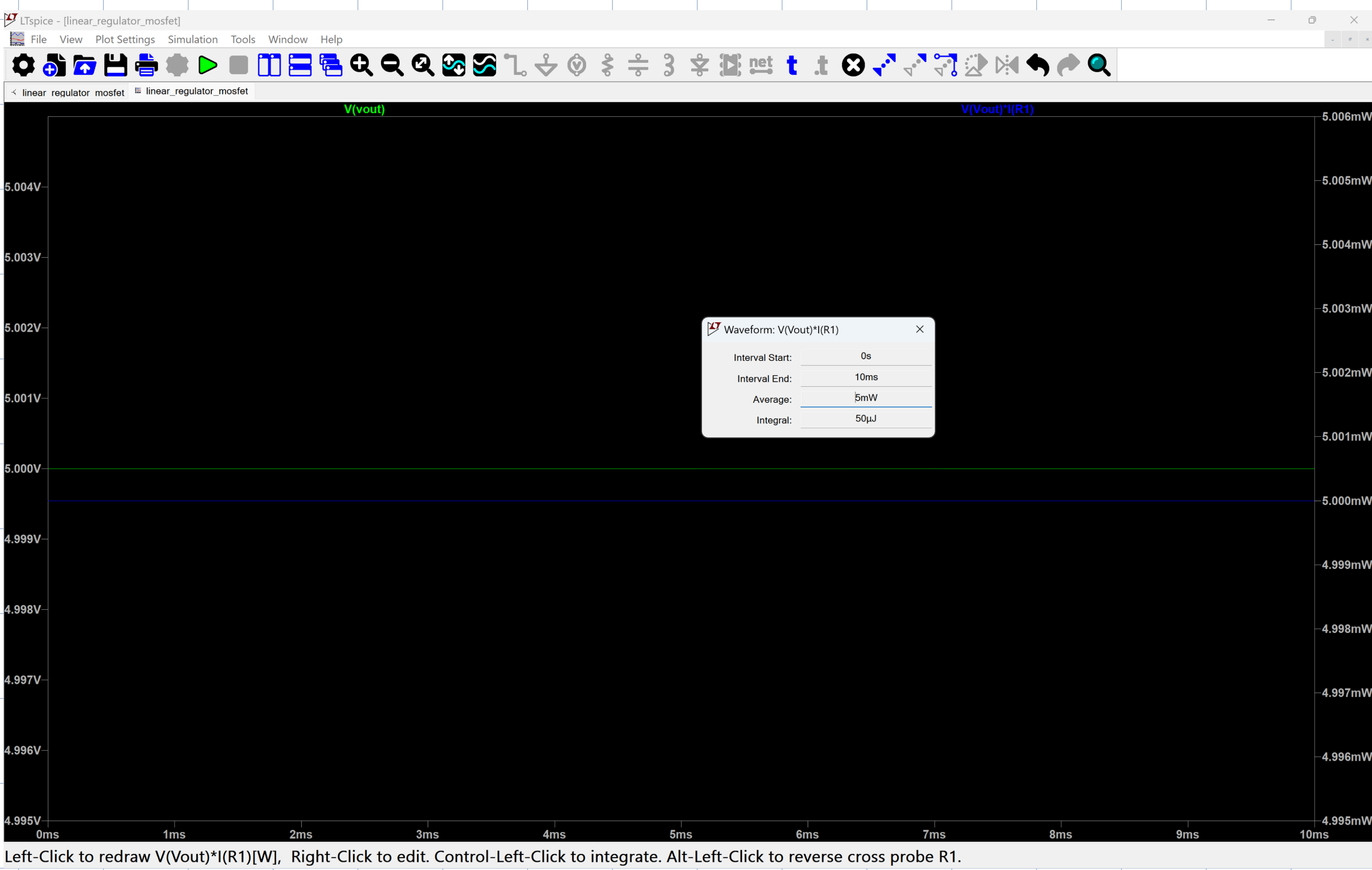
$V_{DS} = V_{GS} = 7V$

$$V_{out} = V_{in} - V_{DS}$$

$$= 12 - 7$$

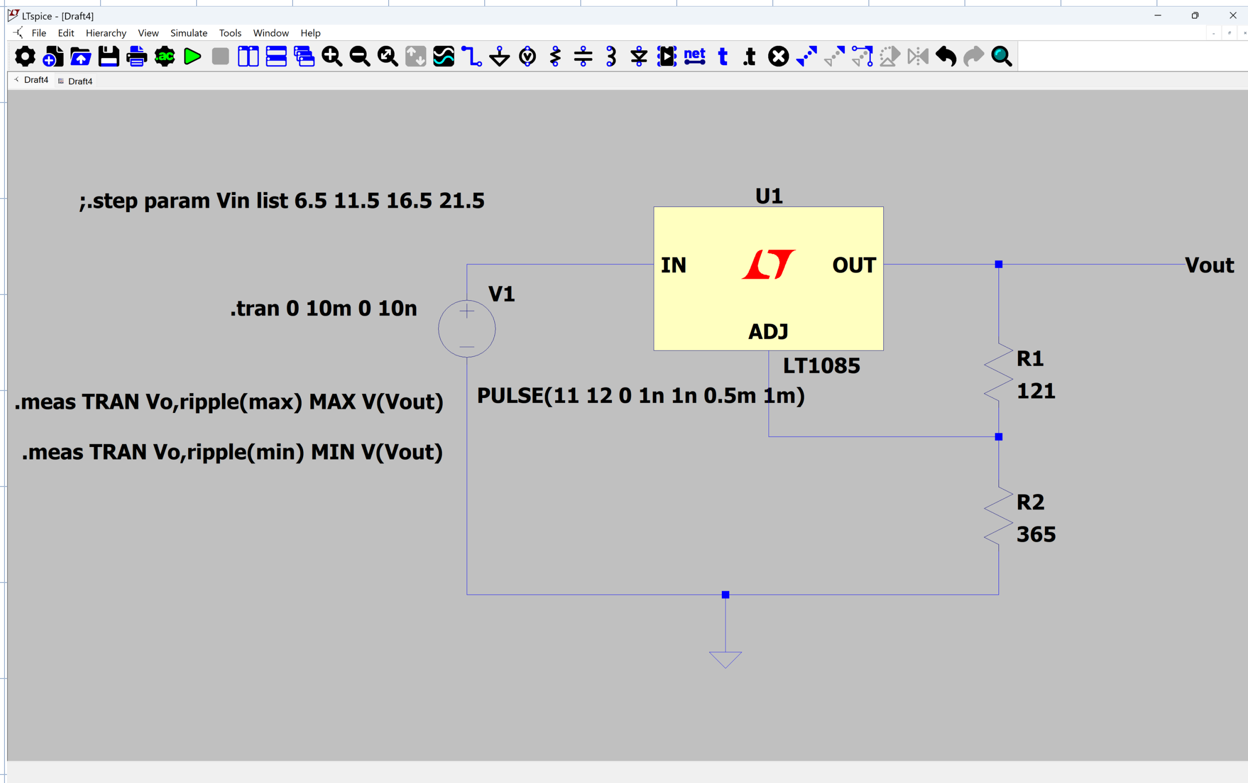
$$\boxed{V_{out} = 5V}$$

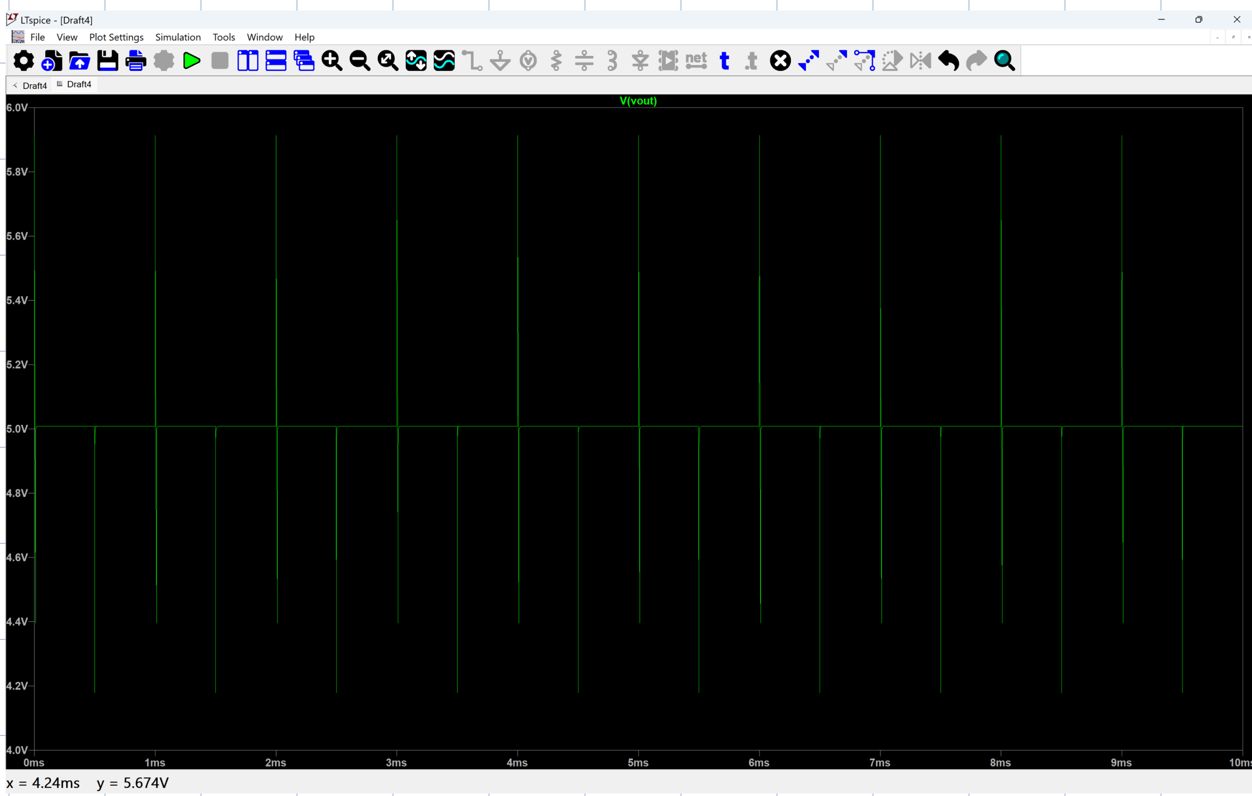

LTspice - [linear_regulator_mosfet]
File Edit Hierarchy View Simulate Tools Window Help
.tran 0 10m 0 10n
D
.model NMOS NMOS(kp=1m vt0=1)
G
M1
NMOS
l=18m w=1m
V1
S
12
Vout
R1
5k
Right click to edit "l=18m w=1m", the Value2 of M1

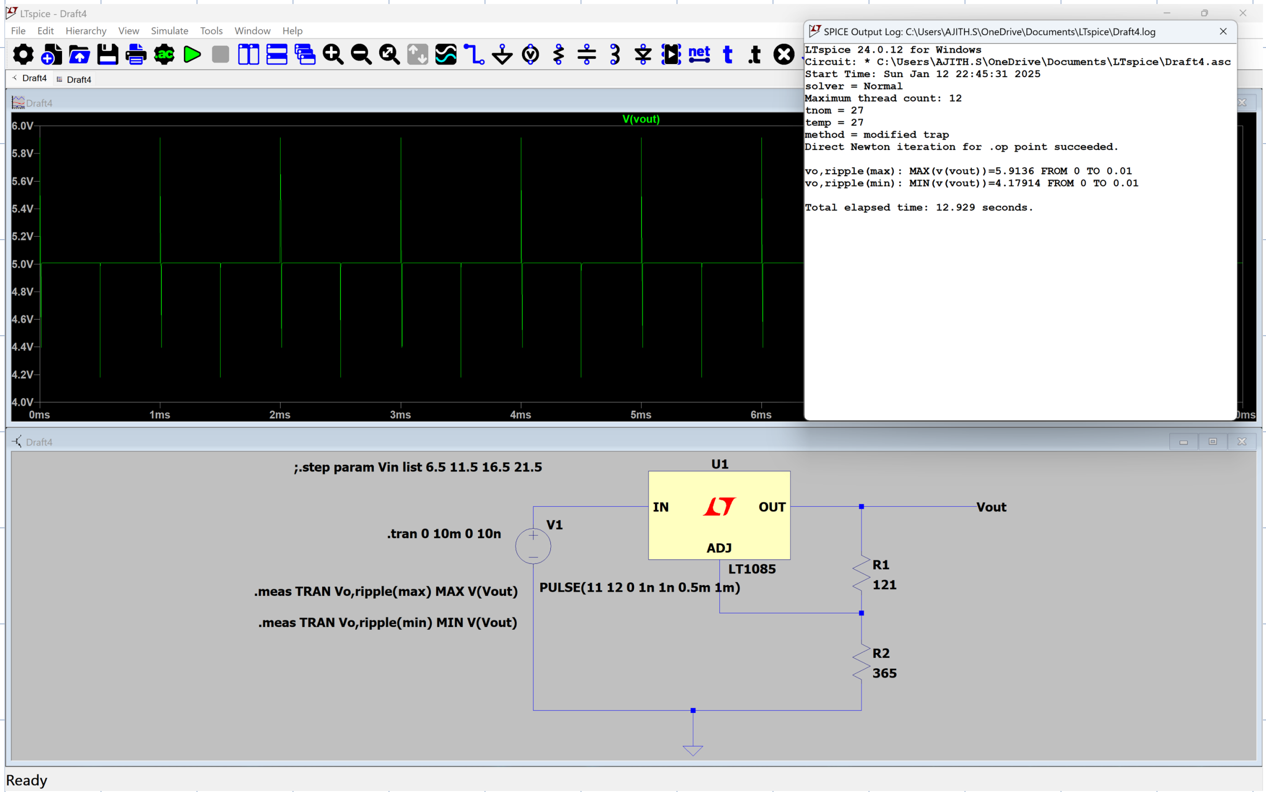

LTspice - [linear_regulator_mosfet]
File View Plot Settings Simulation Tools Window Help
linear_regulator_mosfet
V(vout)
V(Vout)*I(R1)
5.004V
5.003V
5.002V
5.001V
5.000V
4.999V
4.998V
4.997V
4.996V
4.995V
5.006mW
5.005mW
5.004mW
5.003mW
5.002mW
5.001mW
5.000mW
4.999mW
4.998mW
4.997mW
4.996mW
4.995mW
Waveform: V(Vout)*I(R1)
Interval Start: 0s
Interval End: 10ms
Average: 5mW
Integral: 50µJ
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
Left-Click to redraw V(Vout)*I(R1)[W], Right-Click to edit. Control-Left-Click to integrate. Alt-Left-Click to reverse cross probe R1.

Question3

i)

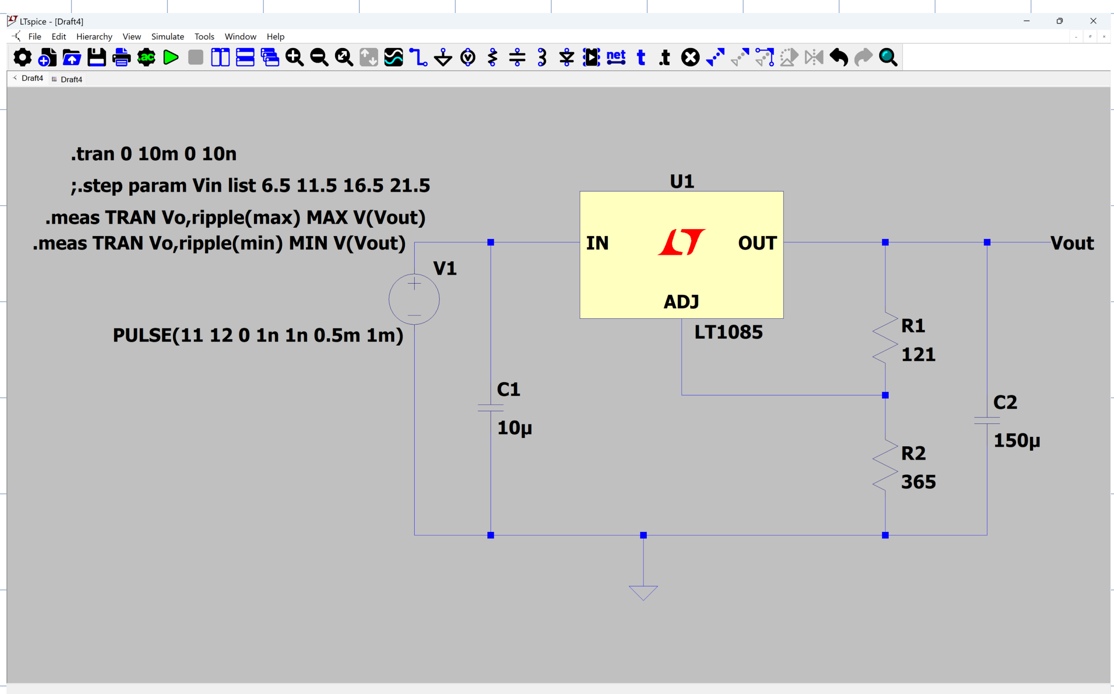

LTspice - [Draft4]
File Edit Hierarchy View Simulate Tools Window Help
Draft4
;.step param Vin list 6.5 11.5 16.5 21.5
U1
IN
OUT
Vout
V1
.tran 0 10m 0 10n
ADJ
LT1085
R1
121
.meas TRAN Vo,ripple(max) MAX V(Vout)
PULSE(11 12 0 1n 1n 0.5m 1m)
.meas TRAN Vo,ripple(min) MIN V(Vout)
R2
365

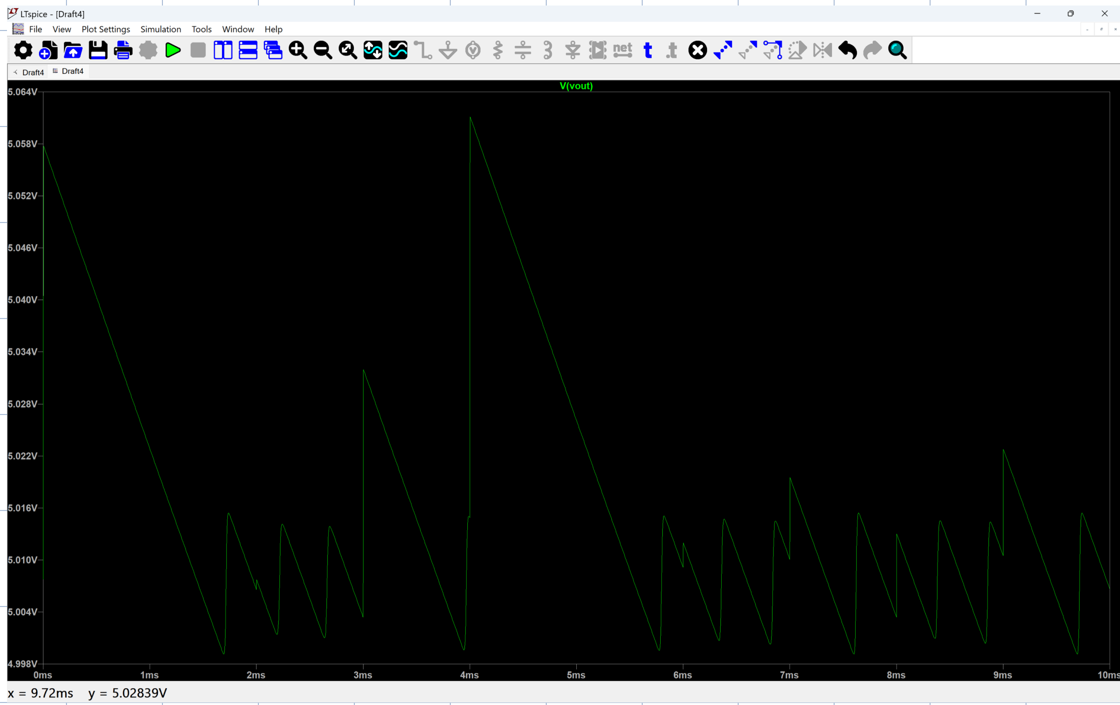

LTspice - [Draft4]
File View Plot Settings Simulation Tools Window Help
Draft4
V(vout)
6.0V
5.8V
5.6V
5.4V
5.2V
5.0V
4.8V
4.6V
4.4V
4.2V
4.0V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
x = 4.24ms y = 5.674V

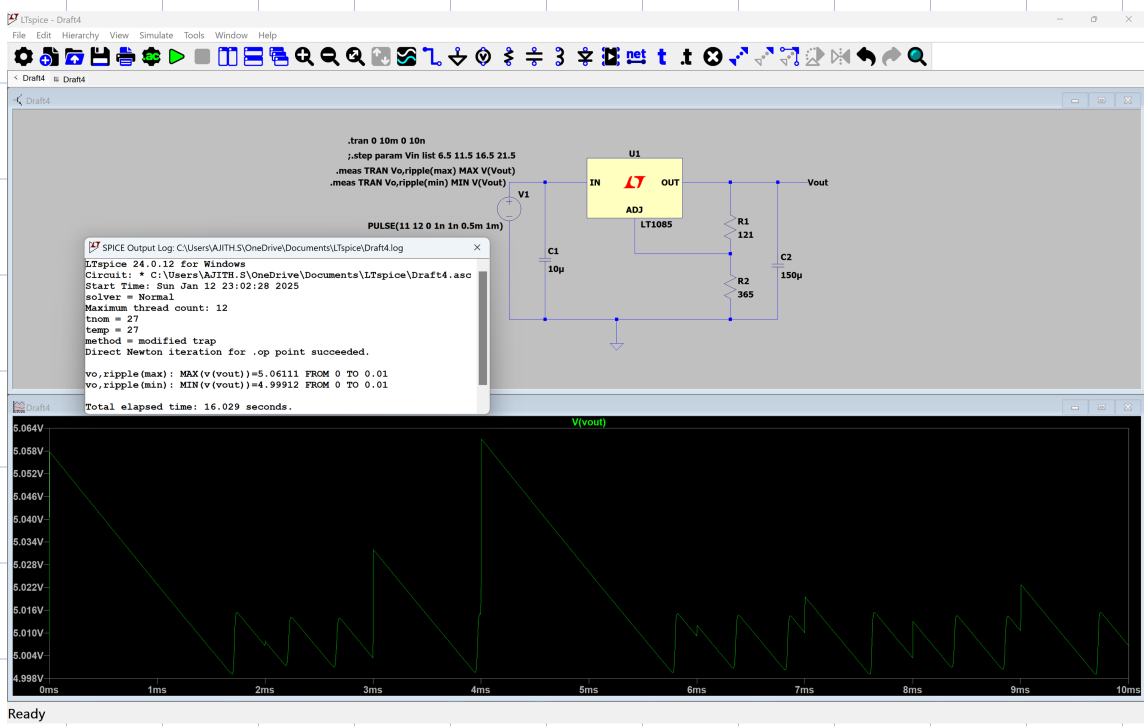


Output ripple max	5.9136 V
output ripple min	4.17914 V
peak to peak values (max - min voltage)	1.73446 V

j)







Output ripple max	5.06111 V
Output ripple min	4.99912 V
peak to peak values (max - min voltage)	0.06199V

We can thus see that the Capacitor at the output reduces the peak to peak ripple from 1.73446 V to 0.06199V.